Description Logics (DLs)

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What are Description Logics?

DLs focus on the representation of terminological knowledge:

Formalize the basic terminology adopted in the modeled application.

- Knowledge Representation formalisms
- Family of description logics with varying expressivity
- Formalism for declarative description of facts
- Come with powerful reasoning service
- Applied in many practical areas
- Basis for the web ontology language OWL of the semantic web

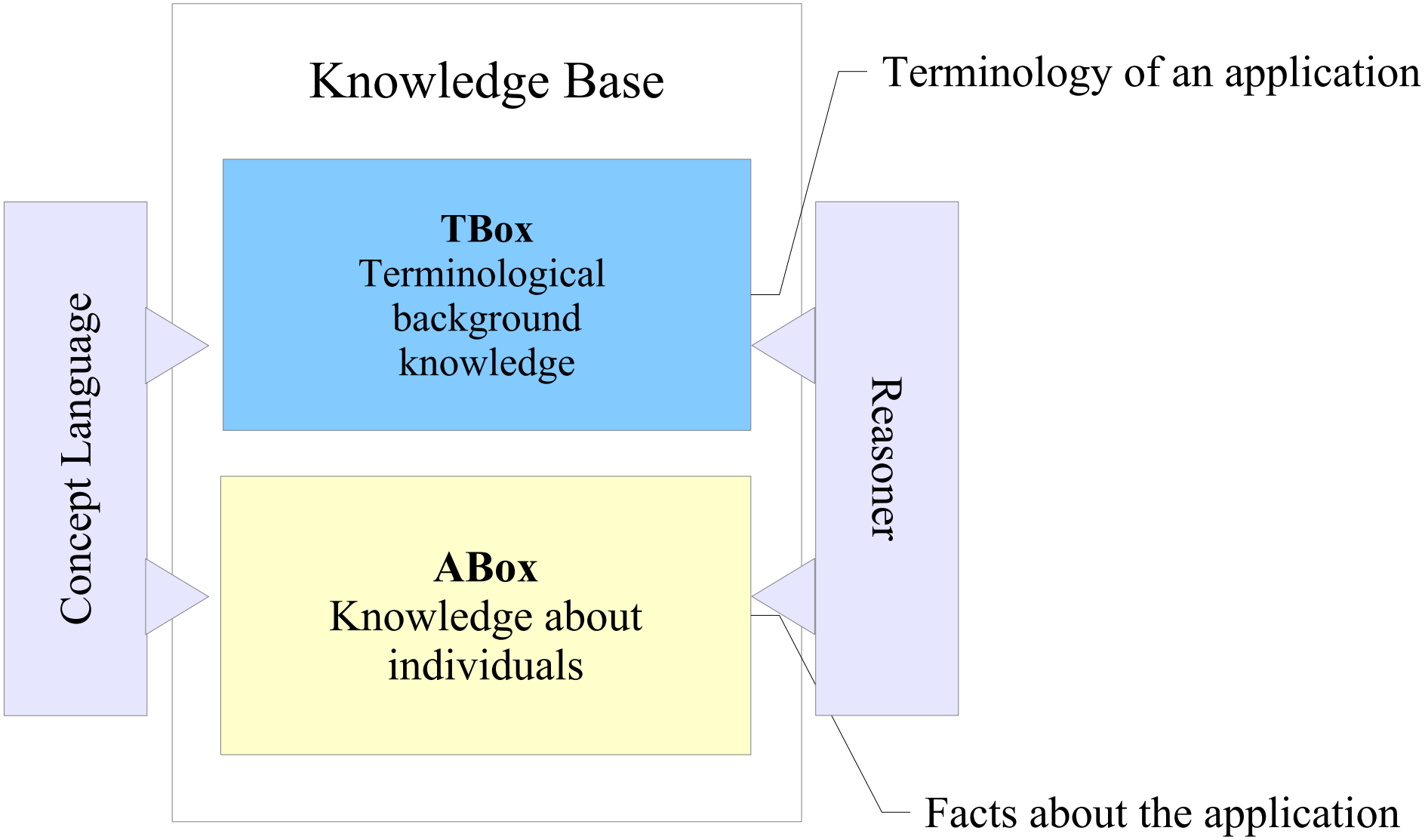
What are DLs used for?

1. Represent terminological knowledge
   a) Characterize concepts in a declarative way
   b) Formal semantics of operators
2. Represent data (assertional knowledge)
   a) Individuals in a declarative way
   b) Formal semantics of assertions
3. Derive new facts
   a) Inferences based on the formal semantics
   b) Find efficient algorithms

Course Outline

1. Terminological Knowledge
   i. Concepts
   ii. TBox
   iii. Terminological reasoning services
2. Assertional Knowledge
   i. Assertions
   ii. ABox
   iii. Assertional reasoning services
3. Reasoning Methods
   i. Reduction
   ii. Tableau
4. DLs and other logics
   i. To FOL
   ii. To ML
5. Conclusions
   i. Layers of research areas

DL Systems


Knowledge Base
Terminology of an application
TBox
Terminological background knowledge
Concept Language
Reasoner
ABox
Knowledge about individuals
Facts about the application

Defining Concepts with DLs

The core part of any DL is the concept language.

$$\mathrm{Mammal} \sqcap \exists \mathrm{hasCover.Fur} \sqcap \forall \mathrm{eats.Meat}$$

- Concept names: assign a name to groups of objects.
- Role names: assign a name to relations between objects.
- Constructors: allow to relate concept names and role names.

Complex concepts can be used in concept definitions.

$$\mathrm{Cat} \equiv \mathrm{Mammal} \sqcap \exists \mathrm{hasCover.Fur} \sqcap \forall \mathrm{eats.Meat}$$

Description Logics: Syntax

Atomic types: concept names $A, B, \cdots \in N_C$
role names $r, s, \cdots \in N_R$

Constructors:		$\mathcal{ALC}$	$\mathcal{EL}$	$\mathcal{FL}_0$
	$\neg C$ (negation)	☺		
	$C \sqcap D$ (conjunction)	☺	☺	☺
	$C \sqcup D$ (disjunction)	☺		
	$\exists r.C$ (existential restriction)	☺	☺	
	$\forall r.C$ (value restriction)	☺		☺
	$\top$ (top concept)	☺	☺	☺
	$\bot$ (bottom concept)	☺		

Example: $\neg(A \sqcup \exists r.(\forall s.B \sqcap \neg A))$.

Semantics of Primitive Concepts

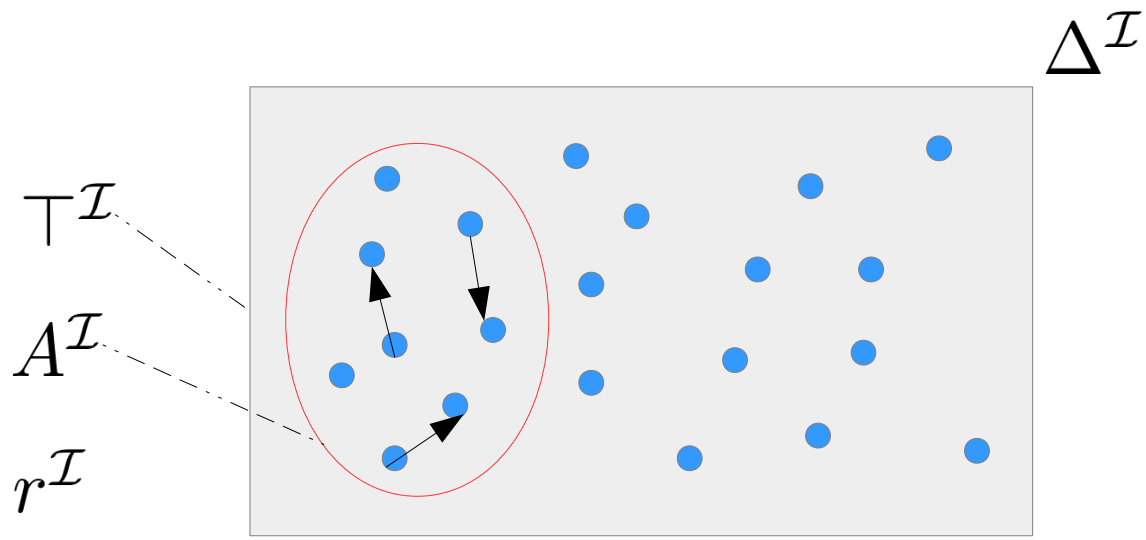
Semantics based on interpretation $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$

Concepts: Subsets of the domain of an interpretation $\Delta^{\mathcal{I}}$

Roles: Binary relations on the domain of an interpretaion $\Delta^{\mathcal{I}}$

Primitive Concepts

$$\top^{\mathcal{I}} = \Delta^{\mathcal{I}}$$
$$\bot^{\mathcal{I}} = \emptyset$$
$$A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$$



Semantics of Complex Concepts

Semantics of complex concepts:

$$(\neg C)^{\mathcal{I}} = \Delta^{\mathcal{I}} \backslash C^{\mathcal{I}}$$

$$(C \sqcap D)^{\mathcal{I}} = C^{\mathcal{I}} \cap D^{\mathcal{I}}$$

$$(C \sqcup D)^{\mathcal{I}} = C^{\mathcal{I}} \cup D^{\mathcal{I}}$$

$$(\exists r.C)^{\mathcal{I}} = \{d \in \Delta^{\mathcal{I}} \mid \exists e \in \Delta^{\mathcal{I}} : (d, e) \in r^{\mathcal{I}} \wedge e \in C^{\mathcal{I}}\}$$

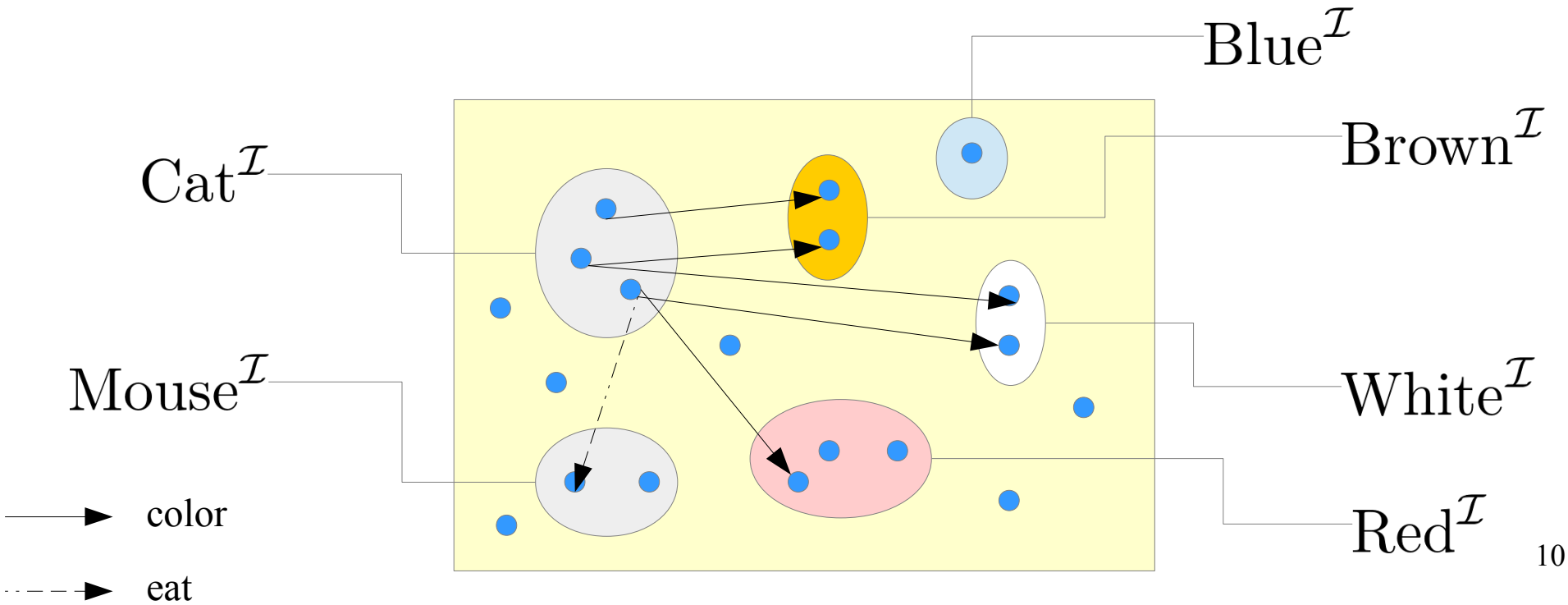
$$(\forall r.C)^{\mathcal{I}} = \{d \in \Delta^{\mathcal{I}} \mid \forall e \in \Delta^{\mathcal{I}} : (d, e) \in r^{\mathcal{I}} \rightarrow e \in C^{\mathcal{I}}\}$$

Example: Concept & Its Model

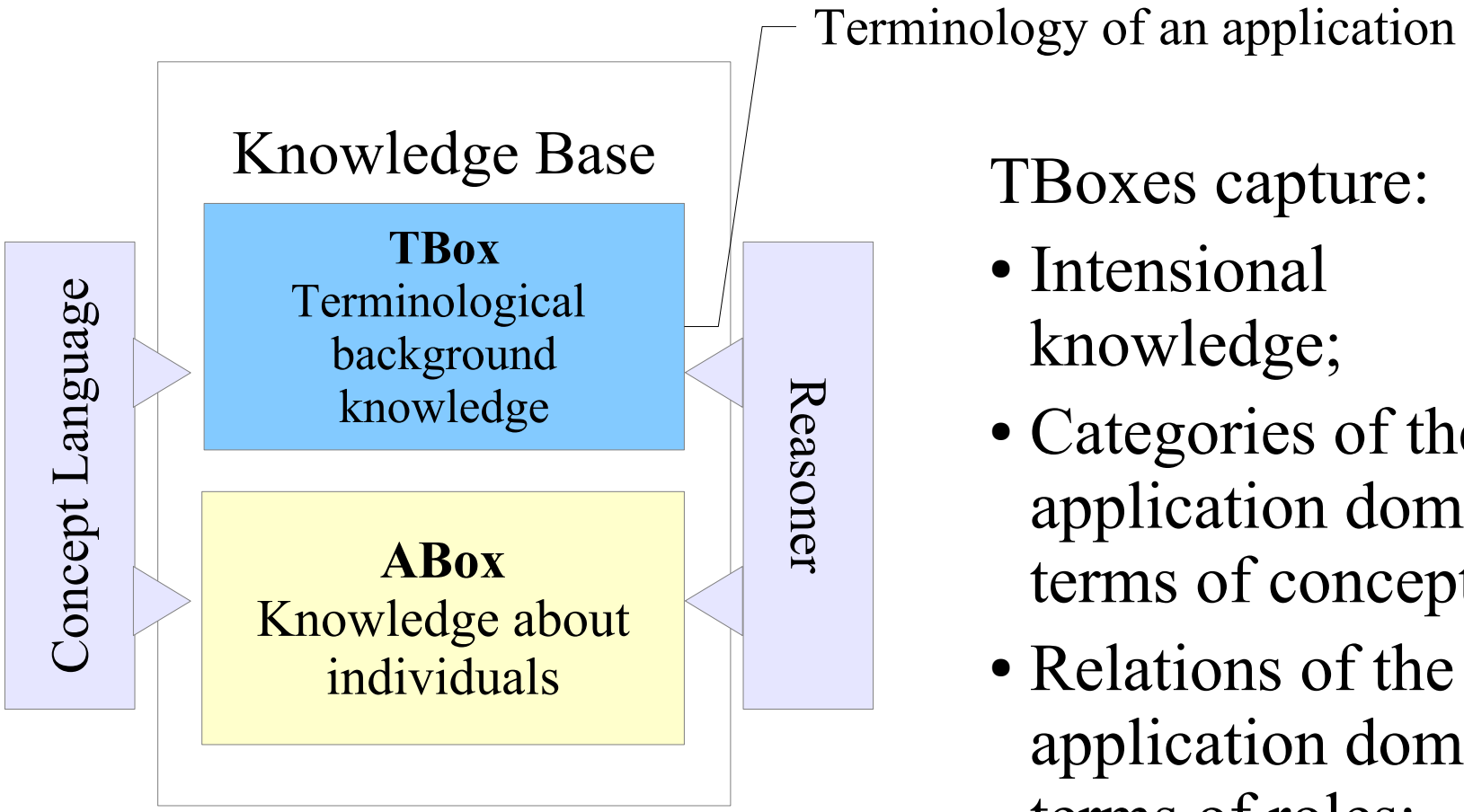
model of a concept: interpretation $\mathcal{I}$ that satisfies the concept.

Example:

$$(\text{Cat} \sqcup \text{Dog}) \sqcap \exists \text{color}.(\text{Brown} \sqcup \text{Black} \sqcup \text{White}) \sqcap \forall \text{color}.\neg\text{Blue}$$



DL systems are more than a concept language



TBoxes capture:

- Intensional knowledge;
- Categories of the application domain in terms of concepts;
- Relations of the application domain in terms of roles;
- Stating super-/sub-concept relationships.

Concept Axioms & TBox

Concept Definitions:

Assign complex concepts to concept names

- Primitive concept definition: $A \sqsubseteq C \quad A \in N_C$;
- (Full) concept definition: $A \equiv C \quad A \in N_C$.

General concept inclusions:

Relate complex concepts to complex concepts

- General concept inclusion (GCI) axiom: $C \sqsubseteq D$;
- General concept equivalence: $C \equiv D$.

TBox $\mathcal{T}$: A finite set of concept axioms.

GCIs can express other concept axioms

- Primitive concept definition: $A \sqsubseteq D \quad A \in N_C$

- Concept definition: $A \equiv D \quad A \in N_C$
  $A \sqsubseteq D$ and $D \sqsubseteq A$

- Concept disjointness axiom: $C \sqcap D \sqsubseteq \bot$

- General concept equivalence: $C \equiv D$
  $C \sqsubseteq D$ and $D \sqsubseteq C$

Kinds of TBoxes

1. A TBox T is a general TBox, if

   - it is a finite set of concept axioms;

   - cyclic definitions and GCIs are allowed.

$$\text{Elephant} \equiv \text{Mammal} \sqcap \exists \text{bodyPart}.\text{Trunk}$$
$$\text{Mammal} \equiv \text{Elephant} \sqcup \text{Lion} \sqcup \text{Zebra}$$

2. A TBox T is an unfoldable TBox, if it has:

   - only (primitive) concept definitions;

   - concept names at most once on the left-hand side of definitions (*uniqueness*).

   - no cyclic definitions, no GCIs.

Semantics of TBox

Semantics of concept axioms

$A \equiv C$ holds in an interpretation $\mathcal{I}$ iff $A^{\mathcal{I}} = C^{\mathcal{I}}$
$C \equiv D$ holds in an interpretation $\mathcal{I}$ iff $C^{\mathcal{I}} \subseteq D^{\mathcal{I}}$

Semantics of TBox

An interpretation $\mathcal{I}$ is a model of a TBox $\mathcal{T}$ if
$C^{\mathcal{I}} = D^{\mathcal{I}}$ for all $C \equiv D \in \mathcal{T}$ and
$C^{\mathcal{I}} \subseteq D^{\mathcal{I}}$ for all $C \sqsubseteq D \in \mathcal{T}$.

Terminological Reasoning Services

1. Concept satisfiability w.r.t. A TBox

Definition. Concept C is satisfiable w.r.t. TBox $\mathcal{T}$ if there is a model $\mathcal{I}$ of $\mathcal{T}$ s.t. $C^{\mathcal{I}} \neq \emptyset$.

- Concepts are intended to represent classes of objects.
- They should not be interpreted as the empty set $\emptyset$.
- Does C state a contradiction w.r.t. $\mathcal{T}$?

Q: $\bot$, Female $\sqcap$ Male, $A \sqcup \neg A$, $A \sqcap \neg A$, $\forall r.\bot$, $\exists r.\top$?

Terminological Reasoning Services

2. Concept subsumption w.r.t. TBox

Definition. Concept C is subsumed by D w.r.t. a TBox $\mathcal{T}$ ($C \sqsubseteq_{\mathcal{T}} D$) if $C^{\mathcal{I}} \subseteq D^{\mathcal{I}}$ for all models $\mathcal{I}$ of $\mathcal{T}$.

Q: $\bot \sqsubseteq A$, $A \sqsubseteq \top$, $A \sqcap B \sqsubseteq A$, $A \sqsubseteq A \sqcup B$?

Terminological Reasoning Services

3. Concept equivalence w.r.t. TBox $(C \equiv_{\mathcal{T}} D)$

   $C \equiv_{\mathcal{T}} D$ iff $C \sqsubseteq_{\mathcal{T}} D$ and $D \sqsubseteq_{\mathcal{T}} C$

4. Classification of the TBox

   Computation of all subsumption relationships between all concept names in $\mathcal{T}$.

   In a naive way, subsumption can be used to compute a concept hierarchy.

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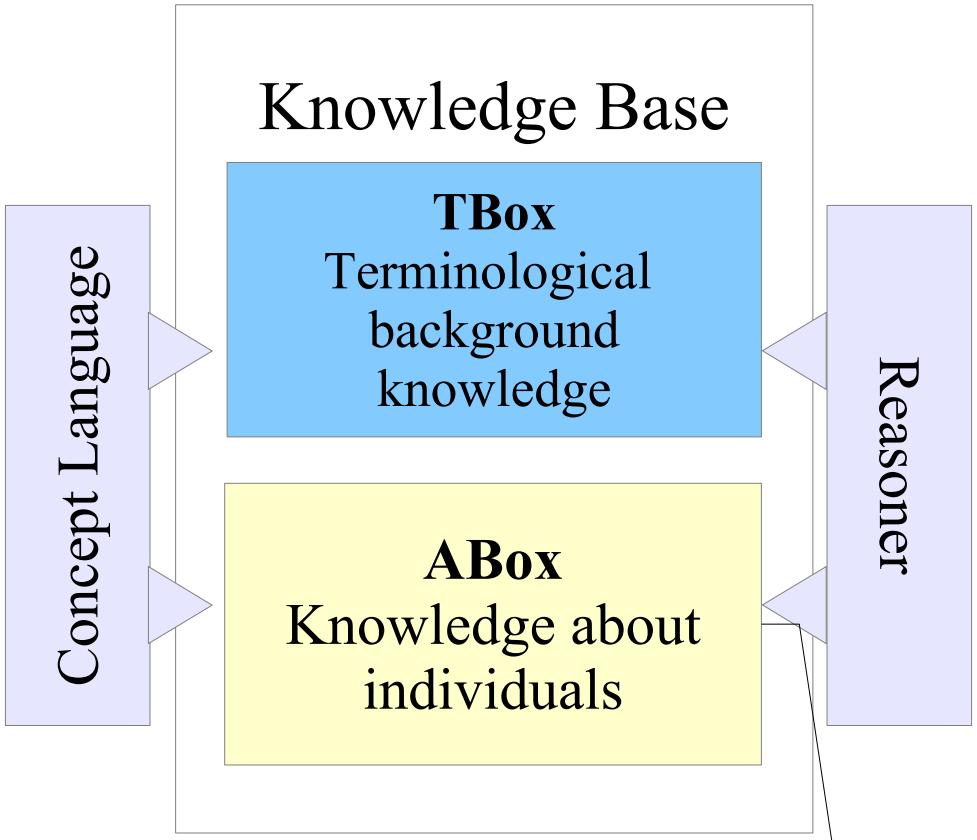

ABoxes capture:

- Extensional knowledge;
- Facts from the application domain;
- Knowledge about individuals;
- Knowledge about the relations between individuals.

Facts about the application

ABox assertions

ABox assertions are:

- Concept assertions: $C(a)$
- Role assertions: $r(a, b)$

Typically, ABoxes use concepts defined in the Tbox.

An interpretation $\mathcal{I}$ maps each individual name $a \in N_I$ to an element of $\Delta^{\mathcal{I}}$.

Semantics of assertions:

- Concept assertions: I satisfies $C(a) \Leftrightarrow a^{\mathcal{I}} \in C^{\mathcal{I}}$; and
- Role assertions: I satisfies $r(a, b) \Leftrightarrow (a^{\mathcal{I}}, b^{\mathcal{I}}) \in r^{\mathcal{I}}$.

ABox & Knowledge Base

An ABox is a partial description of the world.

An ABox A is a finite set of assertions.
A Knowledge Base is a pair $(\mathcal{T}, \mathcal{A})$.

Semantics of ABox:

$\mathcal{I}$ is a model of an ABox $\mathcal{A}$ iff $\mathcal{I}$ satisfies all axioms in $\mathcal{A}$.

Semantics of Knowledge Base (KB):

$\mathcal{I}$ is a model of a KB $\mathcal{K} = (\mathcal{T}, \mathcal{A})$ iff $\mathcal{I}$ is a model of both $\mathcal{T}$ and $\mathcal{A}$.

Example: Assertion

ABox $\mathcal{A}$

Mammal(garfield)	Fur(f17)
Lasagna(I23)	hasCover(garfield, f17)
eats(garfield, I23)	likeMost(garfield, garfield)
$\forall$eats.Beef(garfield)	

Assertional Reasoning Services

1.Knowledge base consistency

Given a KB $\mathcal{K} = (\mathcal{T}, \mathcal{A})$, does $\mathcal{K}$ have a model?

2.Instance checking

Given a KB $\mathcal{K} = (\mathcal{T}, \mathcal{A})$, an individual name a, and a concept C, does $a^{\mathcal{I}} \in C^{\mathcal{I}}$ hold in all models of $\mathcal{K}$?

3.Instance query (a.k.a. Instance retrieval)

Given a KB $\mathcal{K} = (\mathcal{T}, \mathcal{A})$ and a concept C, retrieve all a from $\mathcal{A}$ such that $a^{\mathcal{I}} \in C^{\mathcal{I}}$ hold in all models of $\mathcal{K}$.

4.ABox realization

Given a KB $\mathcal{K} = (\mathcal{T}, \mathcal{A})$, perform an instance query for each named concept in $\mathcal{T}$.

Example: ABox Reasoning

$\mathcal{A}$

Mammal(garfield)	Fur(f17)
Lasagna(I23)	hasCover(garfield, f17)
eats(garfield, I23)	likeMost(garfield, garfield)
$\forall$eats.Beef(garfield)	

$\mathcal{T}$

$$\begin{aligned} \text{Cat} &\equiv \text{Mammal} \sqcap \exists \text{hasCover.Fur} \sqcap \forall \text{eats.Meat} \\ \text{Meat} &\equiv \text{Beef} \sqcup \text{Chicken} \\ \bot &\equiv \text{Lasagna} \sqcap \text{Beef} \end{aligned}$$

- $\mathcal{A}$ is inconsistent w.r.t. $\mathcal{T}$;
- garfield is an instance of Cat.

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Why Reasoning?

TBox and ABox capture implicit information.

Does my knowledge base ...

- Contain a concept that cannot have instances? (since its definition is contradictory)

  Satisfiability w.r.t. TBox checking

- Contain an unwanted synonym for a concept? (unwanted/unindented redundancy in a TBox)

  Concept equivalence checking

- Yield the concept hierarchy I wanted?

  Classification

- Contain individuals not compliant with the specification of the concepts they belong to?

  KB consistency checking

Automated Reasoning

Requirements for good reasoning algorithms.

They should be decision procedures, i.e.

- Terminating;
- Sound; and
- Complete.

We always get an answer.

Every positive answer is correct.

Every negative answer is correct.

Some Good Observations

- $C \sqsubseteq D$ iff $C \sqcap \neg D$ is not satisfiable;
- C is satisfiable if not $C \sqsubseteq \bot$.

- Subsumption can be reduced to (un)satisfiability and vice versa;
- Many reasoning can be reduced to satisfiability (if negation is present in the DL).
- Use reduction and implement one reasoning method.

Reasoning method for unfoldable $\mathcal{ALC}$

1. Use the reduction to reformulate the reasoning problem;
2. Expand concepts w.r.t. TBox;
3. Normalize concepts; and
4. Apply Tableau rules.

Step 1: Reduction

Reformulate a ... as an ABox consistency checking !

1. Satisfiability(C)? $\Longrightarrow$ Consistent: $(\mathcal{T}, \{C(a)\})$?
2. $C \sqsubseteq_{\mathcal{T}} D \Longrightarrow$ Inconsisent: $(\mathcal{T}, \{(C \sqcap \neg D)(a)\})$?
3. $\mathcal{T}, \mathcal{A} \models C(a)$? $\Longrightarrow$ Inconsistent: $(\mathcal{T}, \mathcal{A} \cup \{\neg C(a)\})$?

Implement only consistency checking !

Step 2: Expansion of Concepts

For a naive approach:

Let C be a concept and $\mathcal{T}$ be an unfoldable TBox.

1. replace every concept name of a defined concept with the right-hand side of its definitions $A \equiv C$.

2. repeat until no more replacements can made.

Step 2: Expansion of Concepts

Expansion process terminates due to acyclicity of concept definitions. However, exponential blow-up in the worst case !

$$
\begin{aligned}
\mathcal{T} = \{ & \\
A_0 &\equiv \forall r.A_1 \sqcap \forall s.A_1 \\
A_1 &\equiv \forall r.A_2 \sqcap \forall s.A_2 \\
&\vdots \\
A_{k-1} &\equiv \forall r.A_k \sqcap \forall s.A_k \\
\} &
\end{aligned}
$$

Step 3: Normalize Concepts

A concept C is in negation normal form (NNF) if negation occurs only in front of concept names.

Transformation rules:

$$
\begin{array}{rcl}
\neg\neg C & \rightsquigarrow & C \\
\neg(C \sqcap D) & \rightsquigarrow & \neg C \sqcup \neg D \\
\neg(C \sqcup D) & \rightsquigarrow & \neg C \neg D \\
\neg(\exists r.C) & \rightsquigarrow & \forall r.\neg C \\
\neg(\forall r.C) & \rightsquigarrow & \exists r.\neg C
\end{array}
$$

Step 4: Tableau (Idea Sketch)

Try to construct a model for the input concept C_0 (expanded and in NNF). Represent "candidate models" by a set of proof ABoxes $\mathcal{S}$.

- To decide satisfiability of C_0, start with one initial proof ABox: $A_0 = \{C_0(x_0)\}$ and $\mathcal{S} = \{A_0\}$.
- Apply tableau rules to set of proof ABoxes $\mathcal{S}$ until
  - a proof ABox is complete, i.e. no more rules applicable.
  - there exists an individual x in $\mathcal{A}$ such that $\bot(x) \in \mathcal{A}$ or $\{B(x), \neg B(x)\} \subseteq \mathcal{A}$ for some concept name B. (clash)
- Return "satisfiable" iff a complete and clash-free proof ABox was found (i.e. all proof ABoxes contain a contradiction, return "not satisfiable".)

Tableau Rules

	Precondition	Replace $\mathcal{A}$ by:
$\longrightarrow_{\sqcap}$	$(C_1 \sqcap C_2)(x) \in \mathcal{A}$ $C_1(x) \notin \mathcal{A}$ or $C_2(x) \notin \mathcal{A}$	$\mathcal{A}' := \mathcal{A} \cup \{C_1(x), C_2(x)\}$
$\longrightarrow_{\sqcup}$	$(C_1 \sqcup C_2)(x) \in \mathcal{A}$ $C_1(x) \notin \mathcal{A}$ and $C_2(x) \notin \mathcal{A}$	$\mathcal{A}' := \mathcal{A} \cup \{(C_1)(x)\}$ $\mathcal{A}'' := \mathcal{A} \cup \{(C_2)(x)\}$
$\longrightarrow_{\exists}$	$(\exists r.C)(x) \in \mathcal{A}$, but no z in $\mathcal{A}$ s.t. $\{r(x,z), C(z)\} \subseteq \mathcal{A}$	$\mathcal{A}' := \mathcal{A} \cup \{r(x,z), C(z)\}$
$\longrightarrow_{\forall}$	$\{(\forall r.C)(x), r(x,y)\} \subseteq \mathcal{A}$, but $C(y) \notin \mathcal{A}$	$\mathcal{A}' := \mathcal{A} \cup \{C(y)\}$

Soundness and Completeness (Sketch)

Soundness of the procedure:
is shown by local correctness of each tableau rule.

Local correctness:
Let $\mathcal{S}'$ be obtained from $\mathcal{S}$ by the application of a tableau rule. Then, $\mathcal{S}$ is consistent iff $\mathcal{S}'$ is consistent.

Completeness of the procedure:
is obvious from the definition of a clash.

Termination (Sketch)

The algorithm terminates since:

1. depth of the proof ABox is bounded by maximal nesting of quantifiers in $\mathcal{C}_0$.
2. each individual has at most as many:
   - successors as C_0 has number of sub-concepts.
   - concept assertions as C_0 has number of sub-concepts.
3. concepts are never deleted from node labels.

Tableau Rules

	Precondition	Replace $\mathcal{A}$ by:
$\longrightarrow_{\sqcap}$	$(C_1 \sqcap C_2)(x) \in \mathcal{A}$ $C_1(x) \notin \mathcal{A}$ or $C_2(x) \notin \mathcal{A}$	$\mathcal{A}' := \mathcal{A} \cup \{C_1(x), C_2(x)\}$
$\longrightarrow_{\sqcup}$	$(C_1 \sqcup C_2)(x) \in \mathcal{A}$ $C_1(x) \notin \mathcal{A}$ and $C_2(x) \notin \mathcal{A}$	$\mathcal{A}' := \mathcal{A} \cup \{(C_1)(x)\}$ $\mathcal{A}'' := \mathcal{A} \cup \{(C_2)(x)\}$
$\longrightarrow_{\exists}$	$(\exists r.C)(x) \in \mathcal{A}$, but no z in $\mathcal{A}$ s.t. $\{r(x,z), C(z)\} \subseteq \mathcal{A}$	$\mathcal{A}' := \mathcal{A} \cup \{r(x,z), C(z)\}$
$\longrightarrow_{\forall}$	$\{(\forall r.C)(x), r(x,y)\} \subseteq \mathcal{A}$, but $C(y) \notin \mathcal{A}$	$\mathcal{A}' := \mathcal{A} \cup \{C(y)\}$

<u>Example</u> Check if $(\exists r.A) \sqcap (\exists r.B) \sqsubseteq \exists r.(A \sqcap B)$

C

<u>Proof</u> By reduction, we check $\underline{(\exists r.A) \sqcap (\exists r.B) \sqcap \neg \exists r.(A \sqcap B)}$ is unsatisfiable.

Step 1: Expand the concept and change to NNF.

We get $C_0 = (\exists r.A) \sqcap (\exists r.B) \sqcap \forall r.(\neg A \sqcup \neg B)$

Step 2:

$$\mathcal{A}_0 = \{ ((\exists r.A) \sqcap (\exists r.B) \sqcap \forall r.(\neg A \sqcup \neg B))(x_0) \}$$

$\downarrow$ $\rightarrow \sqcap$

$$\mathcal{A}_1 = \mathcal{A}_0 \cup \{ (\exists r.A)(x_0), (\exists r.B)(x_0), \forall r.(\neg A \sqcup \neg B)(x_0) \}$$

$\downarrow$ $\rightarrow \exists$

$$\mathcal{A}_2 = \mathcal{A}_1 \cup \{ r(x_0, x_1), A(x_1) \}$$

$\downarrow$ $\rightarrow \exists$

$$\mathcal{A}_3 = \mathcal{A}_2 \cup \{ r(x_0, x_2), B(x_2) \}$$

$\downarrow$ $\rightarrow \forall$

$$\mathcal{A}_4 = \mathcal{A}_3 \cup \{ (\neg A \sqcup \neg B)(x_1), (\neg A \sqcup \neg B)(x_2) \}$$

$\rightarrow \sqcup$ $\rightarrow \sqcup$

$$\mathcal{A}_5' = \mathcal{A}_4 \cup \{ \neg A(x_1) \} \qquad \mathcal{A}_5'' = \mathcal{A}_4 \cup \{ \neg B(x_1) \} \qquad \mathcal{A}_5''' = \mathcal{A}_4 \cup \{ \neg A(x_2) \}$$

$\times$ $\bigcirc$ $\bigcirc$

$$\mathcal{A}_5'''' = \mathcal{A}_4 \cup \{ \neg B(x_2) \}$$

$\times$

Step 3: Since C_0 is satisfiable, then $(\exists r.A) \sqcap (\exists r.B) \not\sqsubseteq \exists r.(A \sqcap B)$.

A witness: $\Delta^{\mathcal{I}} = \{x_0, x_1, x_2\}$ $\quad r^{\mathcal{I}} = \{(x_0, x_1), (x_0, x_2)\}$

$A^{\mathcal{I}} = \{x_1\}$ $\quad B^{\mathcal{I}} = \{x_2\}$

With this interpretation, $x_0 \in C_0^{\mathcal{I}}$, i.e. $x_0 \in ((\exists r.A) \sqcap (\exists r.B))^{\mathcal{I}}$

but $x_0 \notin (\exists r.(A \sqcap B))^{\mathcal{I}}$

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DLs and FOL

Basic Correspondence:

concept names A	$\Longleftrightarrow$	unary predicates P_A
role names r	$\Longleftrightarrow$	binary predicates P_r
concepts	$\Longleftrightarrow$	formulas with one free variable
individuals	$\Longleftrightarrow$	constants c_a

Obvious Translation

$$
\begin{aligned}
\varphi^x(A) &= P_A(x) \\
\varphi^x(\neg C) &= \neg\varphi^x(C) \\
\varphi^x(C \sqcap D) &= \varphi^x(C) \wedge \varphi^x(D) \\
\varphi^x(C \sqcup D) &= \varphi^x(C) \vee \varphi^x(D) \\
\varphi^x(\exists r.C) &= \exists y.P_r(x,y) \wedge \varphi^y(C) \\
\varphi^x(\forall r.C) &= \forall y.P_r(x,y) \rightarrow \varphi^y(C)
\end{aligned}
$$

<u>Remark:</u> Not all DLs can be represented By FOL (e.g. transitive closure).

DLs and ML

Obvious Translation:

concept names	$\Longleftrightarrow$	propositional variables
role names	$\Longleftrightarrow$	modal parameters
concepts $\exists r.C$	$\Longleftrightarrow$	formulas $\Diamond\varphi$
concepts $\forall r.C$	$\Longleftrightarrow$	formulas $\Box\varphi$

Remark:

- Interpretations can be viewed as Kripke semantics.
- $\mathcal{ALC}$ is a syntactical variant of modal K_ω.
- Extensions of $\mathcal{ALC}$ are related to graded modal logic, PDL, etc.

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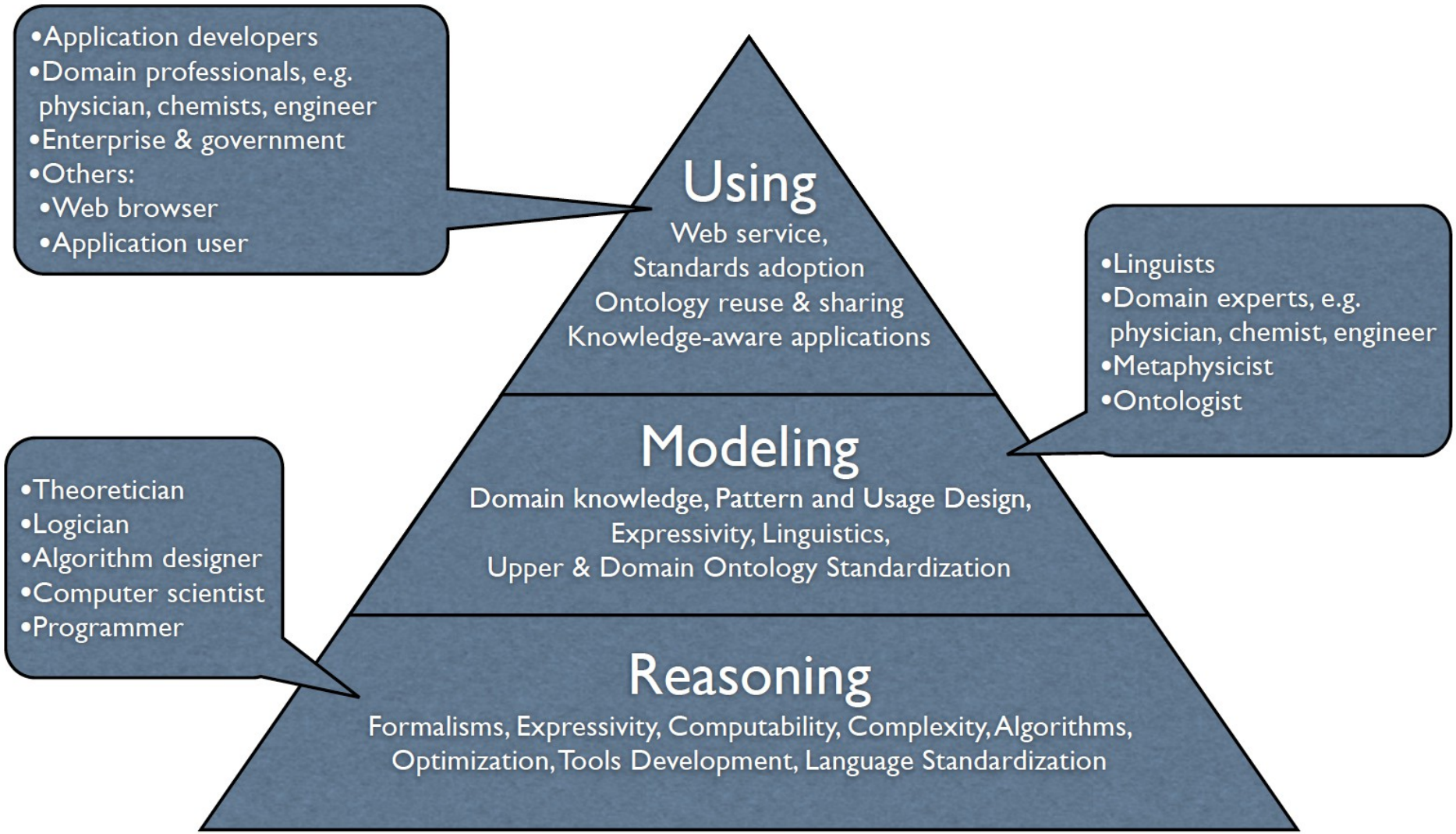
Layers of Research Area



Web Ontology Language

- Web Ontology Language (OWL) is a W3C recommendation as an ontology language for the Semantic Web.
  - Conforms the layer architecture of Web standards.
  - Reuses RDF vocabulary, the XML syntax & namespace, and the Web unicode and URI.
  - **Has well-defined syntax and semantics based on DLs.**
  - Supported by several tools (both editing and reasoning)
- Current version is OWL 2.